

Eran Rehani

PHYSICS STUDENT · HEBREW UNIVERSITY OF JERUSALEM

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Physics student at the Hebrew University of Jerusalem, completing a B.Sc. in Physics (Extended Program) in 2027. I work in computational physics and scientific programming, building numerical methods from first principles (Monte Carlo, event-driven molecular dynamics, numerical quantum mechanics) and validating them against exact results: Onsager's critical coupling, the analytic hydrogen spectrum, Maxwell–Boltzmann statistics.

EDUCATION

B.Sc. Physics (Extended Program) · Hebrew University of Jerusalem 2023–
Year 3 complete, remaining coursework in progress; degree expected 2027. $\LaTeX$ summaries of the Year 3 courses published at [lecture-notes-huji](#).

EXPERIENCE

Personal Trainer · Certified, Wingate Institute 2021–present
Certified Personal Trainer, Wingate Institute. Built training programmes, tracked client progress, and managed my own client book, first at a gym and then independently.
Still training one shift a week alongside full-time studies.

MILITARY SERVICE

Intelligence Corps, IDF · Cyber & Information Security 12/2018–5/2021
Worked in Linux and networking environments; wrote Python and Bash tooling to automate recurring analysis. Owned a technical domain end to end and trained incoming personnel on it.
Handled live security incidents on an on-call rotation, under time pressure and with sensitive data.

IAF Pilot Course · Ben-Gurion University 7/2018–12/2018
Passed national Israeli Air Force screening and entered the pilot course; completed the academic coursework Ben-Gurion University delivers to the cohort.

TECHNICAL SKILLS

Programming	Python, MATLAB, $\LaTeX$
Libraries	NumPy, SciPy, Matplotlib, Numba (JIT), OpenCV, multiprocessing
Tools	Git, Linux, Jupyter, VS Code, AI-assisted development workflows
Languages	Hebrew (native), English (high level, university exemption); writes technical reports in English

RELEVANT COURSEWORK

Physics: Analytical Electrodynamics, Analytical Mechanics, Quantum Mechanics 1–2, Statistical Physics, Approximation Methods, Nuclear Physics, Particle Physics, Continuum Mechanics, Waves & Optics, Thermal Physics, Labs A–B. **Mathematics:** Linear Algebra, Calculus, Probability, Complex Functions, Equations of Mathematical Physics. **Computational:** Computational Physics (Python), MATLAB.

PROJECTS & LAB WORK

Chaos & Nonlinear Dynamics · Python · NumPy · Matplotlib 2024
Period-doubling cascade of the logistic map with cobweb plots; extracted the Feigenbaum constant $\delta \approx 4.669$. Poincaré sections, phase portraits and Lyapunov exponents from driven RLD circuit lab data. Lab reports: [\[drip faucet\]](#) [\[RLD\]](#)

Monte Carlo & Molecular Dynamics Simulations · Python · Numba · multiprocessing 2025–2026
2D Ising model, heat-bath (Glauber) dynamics under Numba JIT. Reproduced Onsager's exact critical coupling $\eta_c = \ln(1 + \sqrt{2})/2$ and extracted β, γ, ν by finite-size scaling with the Binder cumulant.
Hard-Disk Gas MD: event-driven dynamics, collision times solved analytically (no time-step error). Verified Maxwell–Boltzmann velocities and computed the pair correlation $g(r)$.

Numerical Schrödinger Solver · Python · NumPy · SciPy 2026
Radial Schrödinger equation by finite differences and Numerov, benchmarked on the harmonic oscillator and hydrogen: Numerov converges as $O(K^{-4})$ against $O(K^{-2})$. Added perturbative corrections and an SCF solver with density mixing, reproducing the helium and oxygen ground states.

Computational Physics (77315) · Python · Jupyter 2026
Floating-point stability, interpolation, numerical integration and root-finding, applied to the structure of white dwarf stars.